

Code:

1. Server.java

```
import java.io.*;
import java.net.*;
import java.util.*;

public class GroupChatServer {
    private static final int PORT = 4568;
    private static final Set<PrintWriter> clientWriters =
Collections.synchronizedSet(new HashSet<>());

    public static void main(String[] args) {
        System.out.println("Group Chat Server is running...");

        try (ServerSocket serverSocket = new ServerSocket(PORT)) {
            while (true) {
                Socket clientSocket = serverSocket.accept();
                System.out.println("New client connected: " +
clientSocket);

                new ClientHandler(clientSocket).start();
            }
        } catch (IOException e) {
            e.printStackTrace();
        }
    }

    private static class ClientHandler extends Thread {
        private Socket socket;
        private PrintWriter output;
        private BufferedReader input;
        private String clientName;

        public ClientHandler(Socket socket) {
            this.socket = socket;
        }

        public void run() {
            try {
```

```

        input = new BufferedReader(new
InputStreamReader(socket.getInputStream()));
        output = new PrintWriter(socket.getOutputStream(), true);

        clientWriters.add(output);

        output.println("Enter your name:");
        clientName = input.readLine();
        broadcast(clientName + " has joined the chat.");

        String message;
        while ((message = input.readLine()) != null) {
            if (message.equalsIgnoreCase("exit")) {
                break;
            }
            broadcast(clientName + ": " + message);
        }

        clientWriters.remove(output);
        broadcast(clientName + " has left the chat.");
        socket.close();
    } catch (IOException e) {
        e.printStackTrace();
    }
}

private void broadcast(String message) {
    synchronized (clientWriters) {
        for (PrintWriter writer : clientWriters) {
            writer.println(message);
        }
    }
    System.out.println(message);
}
}
}

```

2. Client.java

```
import java.io.*;
import java.net.*;
import java.util.Scanner;

public class GroupChatClient {
    public static void main(String[] args) {
        String HOST = "127.0.0.1";
        int PORT = 4568;

        try (Socket socket = new Socket(HOST, PORT);
            BufferedReader input = new BufferedReader(new
InputStreamReader(socket.getInputStream()));
            PrintWriter output = new
PrintWriter(socket.getOutputStream(), true);
            Scanner scanner = new Scanner(System.in)) {

            System.out.print(input.readLine() + " ");
            String name = scanner.nextLine();
            output.println(name);

            new Thread(() -> {
                try {
                    String serverMessage;
                    while ((serverMessage = input.readLine()) != null) {
                        System.out.println(serverMessage);
                    }
                } catch (IOException e) {
                    System.out.println("Disconnected from chat.");
                }
            }).start();

            while (true) {
                String message = scanner.nextLine();
                if (message.equalsIgnoreCase("exit")) {
                    output.println("exit");
                    System.out.println("You left the chat.");
                    break;
                }
                output.println(message);
            }
        }
    }
}
```

```

    } catch (IOException e) {
        e.printStackTrace();
    }
}
}
}

```

Output:

```

PS D:\Github\SEM8_MATERIAL\DC\CODE\DC_EXP2> java -va GroupChatServer.java
PS D:\Github\SEM8_MATERIAL\DC\CODE\DC_EXP2> java -va GroupChatClient
Group Chat Server is running...
New client connected: Socket[addr=/127.0.0.1,port=56203,localport=4568]
Alice has joined the chat.
New client connected: Socket[addr=/127.0.0.1,port=56205,localport=4568]
Bob has joined the chat.
Alice has left the chat.
Bob has left the chat.

PS D:\Github\SEM8_MATERIAL\DC\CODE\DC_EXP2> java -va GroupChatClient
Enter your name: Alice
Alice has joined the chat.
Bob has joined the chat.
exit
You left the chat.
Disconnected from chat.
PS D:\Github\SEM8_MATERIAL\DC\CODE\DC_EXP2>

```